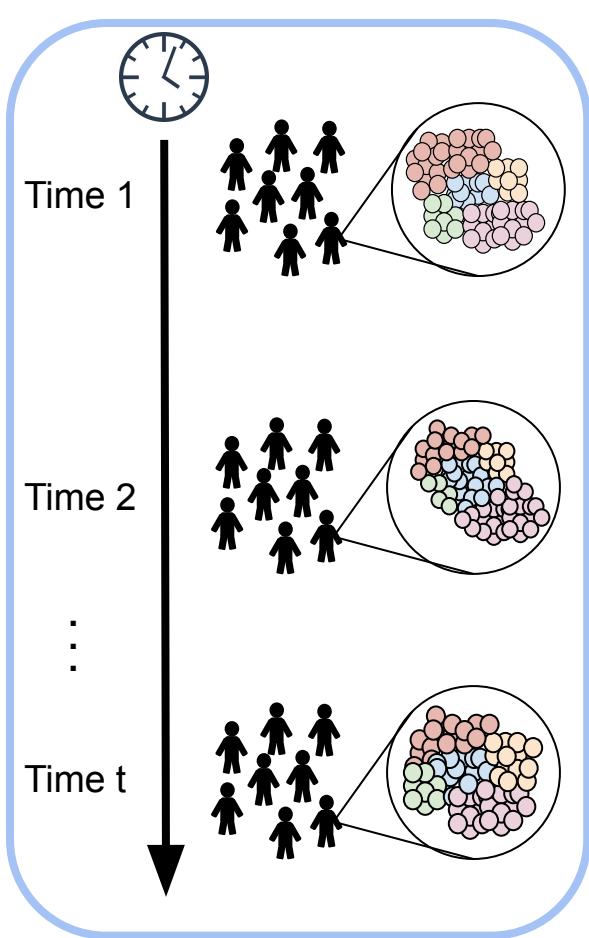
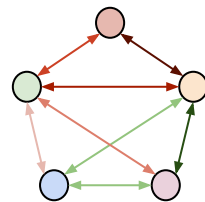




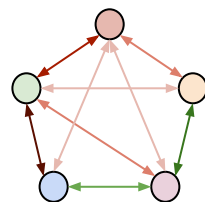
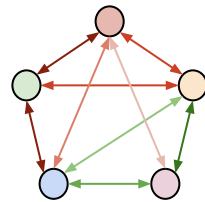
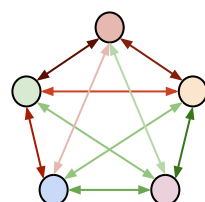
$$\begin{aligned} \# \text{red circle}_t &= a_i^{\text{red}} + b_i^{\text{red}}t + \beta_0^{\text{red}} + \beta_1^{\text{red}}t + \varepsilon_t^{\text{red}} \\ &\vdots \\ \# \text{green circle}_t &= a_i^{\text{green}} + b_i^{\text{green}}t + \beta_0^{\text{green}} + \beta_1^{\text{green}}t + \varepsilon_t^{\text{green}} \end{aligned}$$

$$\begin{matrix} a_i^{\text{red}} \\ b_i^{\text{red}} \\ \vdots \\ a_i^{\text{green}} \\ b_i^{\text{green}} \end{matrix} \sim \begin{matrix} \text{[Matrix of colored squares]} \\ \end{matrix} =: \mathbf{D}$$

Association of Evolution



Evolution of Association



Time 1

Time 2

...

Time t